

Does Protecting the Router in Quantized MoE Improve Accuracy, and How?

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Abstract

Mixture-of-Experts (MoE) models spend almost all of their parameters in expert networks, and only a tiny fraction in the routers that choose them. We ask a narrow question: in calibration-free, weight-only post-training quantization, does leaving just those routers in BF16 protect routing and language-model quality? On OLMoE-1B-7B and Qwen1.5-MoE-A2.7B, at INT8, INT4 and INT3, the answer is yes for routing. Protecting the routers reduces routing KL and top-1 expert flips relative to quantizing them, with disjoint bootstrap intervals. A parameter-count-matched placebo does nothing. We then split the drift that remains between its two possible causes, the error in the activations reaching the router and the rounding of the router’s own weights, to see which contributes more: the activations dominate at INT8, the rounded weights at INT4. Router protection is cheap and specific.

1 Introduction

A Mixture-of-Experts layer sends each token through a small linear *router* and then through only a handful of *experts* (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024). The experts are almost the whole model. The routers are not: 0.030% of OLMoE-1B-7B and 0.021% of Qwen1.5-MoE-A2.7B. Quantizing the experts therefore saves nearly everything. Quantizing the routers as well can send a token to the wrong experts.

We test one safeguard, with no calibration: keep every router in BF16 and quantize every expert. The comparison is a single variable. UNIFORM quantizes routers and experts at the same INT- N width; MIXED quantizes the same experts and leaves the routers untouched; GOLD is the BF16 reference. A parameter-count-matched PLACEBO

then asks whether protecting any equally small set of weights would have done as well.

This is not a new proposal. Fu et al. (2025) already run similar experiments and Fang and Huang (2026) argue that freezing the router is not enough. What those papers do not isolate is the calibration-free, weight-only version of the variable, scored on routing itself, with a control and a mechanism split. That is the measurement we report.

Three findings, previewed in Figure 1. First, MIXED beats UNIFORM on routing KL and top-1 expert flips in every model $\times$ bit-width cell we ran. Second, the gain is specific to the routers: the placebo lands on UNIFORM. Third, we split the drift that remains between the hidden states arriving at the router and the rounding of the router’s own weights (Section 6), and find that the hidden states are the larger source of flips at INT8, while for INT4 the router’s own weights are the main cause.

2 Background

MoE routing. At each MoE layer a linear router maps a hidden state $x \in \mathbb{R}^d$ to E logits. The layer keeps the top- k experts and mixes their outputs with a softmax over those k scores:

$$y = \sum_{i \in \text{Top}k(Wx)} \text{softmax}(Wx)_i E_i(x). \quad (1)$$

A flip of the top-1 index, or of the top- k set, changes which parameters compute y . OLMoE uses $E=64$, $k=8$; Qwen1.5-MoE uses $E=60$, $k=4$, plus a shared expert that we do *not* protect (Section 4).

Post-training quantization. We use weight-only, per-output-channel symmetric INT- N quantization through TorchAO (Or et al., 2025), with scales taken from the weight tensor itself. There is no calibration set, no rotation, and no reconstruction loss.

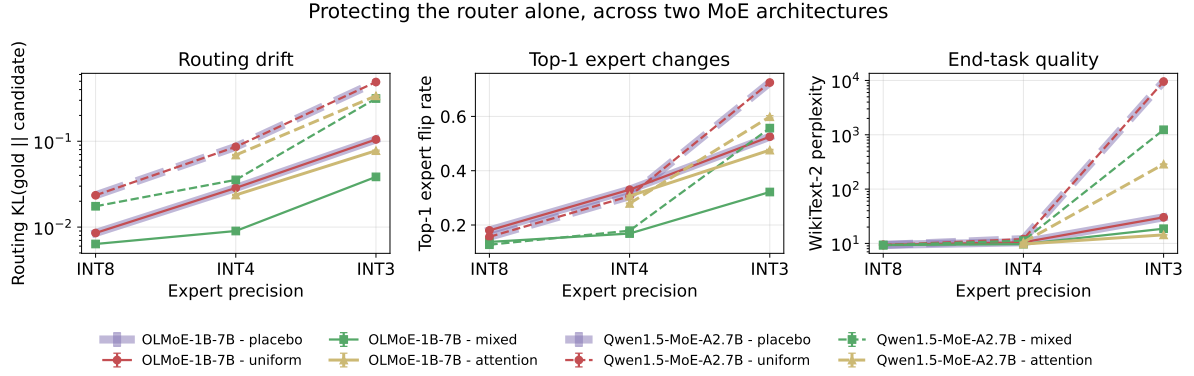


Figure 1: Routing KL (left, log scale), top-1 expert flip rate (centre) and WikiText-2 perplexity (right, log scale) against expert precision. MIXED keeps every router in BF16; UNIFORM quantizes the routers too; PLACEBO protects a parameter-count-matched non-router module instead. Error bars are 95% bootstrap intervals over sequences. PLACEBO sits on UNIFORM, that coincidence is the control result. The extra series, ATTENTION, is a follow-up that also leaves attention in BF16 (Section 5); it is not a substitute for protecting the router.

Embeddings and the language-model head stay in BF16 in every policy. This is a different regime from GPTQ, AWQ, SmoothQuant or LLM.int8() (Frantar et al., 2023; Lin et al., 2024; Xiao et al., 2023; Dettmers et al., 2022), all of which fit something to data.

Metrics. Against the GOLD router we report the KL between full softmaxes (*routing KL*), the rate at which the top-1 expert changes (*top-1 flip*), and the Jaccard distance of the top- k sets. WikiText-2 perplexity (Merity et al., 2017) is a point estimate over every scored token; it carries no confidence interval, and the decision gate does not use it.

3 Related Work

Router precision in MoE quantization. Protecting the router is already standard practice. EAQuant runs experts at W4A4 (both weights and activations at BF4) with the router at W8A8 (Fu et al., 2025). On the same OLMoE checkpoint we use, their Table 4 holds experts at W3A4 and moves the router from W3A4 to W8A8; WikiText-2 perplexity falls by 0.32 and average zero-shot accuracy rises by about one point. That is a real router-precision effect, and we do not claim otherwise. It is also confounded in three ways this paper removes. Every condition moves router *weight and activation* precision together; the baseline is a calibrated rotation method; and the most-protected router is W8A8, never a router exempt from quantization. The table is scored only on end-task metrics, so it cannot say whether routing itself changed. GEMQ independently notes that routers are under 0.04% of MoE parameters (Deng et al., 2026),

which matches our census.

Repairing routing with data. A second line of work treats routing drift as something to fit. VS-RAQ adds value- and structure-alignment losses at 4-bit (Park et al., 2026). Hyeon and Do (2026) argue that retraining-free compression is not enough and that the router needs calibration. Both papers are the opposing regime: they spend data on the router. We spend none.

Whether protection is sufficient. Fang and Huang (2026) argue that freezing the router is insufficient: quantized expert outputs shift the hidden states that reach the next layer’s router, so a protected router still misroutes. Their controlled router-precision table on OLMoE appears to contradict us directly, with an FP16 router slightly *worse* than a quantized one under W4A4 experts. Their Table 8 splits that comparison by whether their alignment losses are active, and on our reading the two branches disagree. Without the losses, the FP16 router wins on OLMoE and DeepSeek-MoE and ties on Qwen3-MoE; router quantization overtakes protection only once a calibration-fitted objective has realigned it. The disagreement is therefore conditional on calibration rather than absolute. We occupy the calibration-free branch, where their own numbers favour protection, and we score it on routing behaviour, which they report only through end-task metrics, not as KL, flip rate, or load collapse.

Bit-width sweeps over MoE sub-structures. QuantMoE-Bench sweeps precision across attention, experts, block depth and linear types *inside*

experts (Li et al., 2024). The router is not a condition. Its “gate projection” analysis is the FFN gate inside each expert, not the routing gate. MoQE found that expert FFNs tolerate much lower precision than attention (Kim et al., 2023). That fact motivates a follow-up in Section 5, not the main design.

4 Method

Three policies, one variable. GOLD is BF16 everywhere. UNIFORM quantizes every linear module at INT- N , including every router. MIXED uses the same INT- N policy and skips the router. Qwen’s shared_expert_gate is left quantized, so MIXED means the same thing on both models. Offline inspection of the stored router weights confirms the single-variable claim: every MIXED router is bit-identical to GOLD in every layer, and every UNIFORM router sits at the INT- N level ceiling at INT4 and INT3, and well below it at INT8 (Appendix B).

A parameter-count-matched control. PLACEBO is UNIFORM except that one randomly chosen non-router module, matched to the total router parameter budget, stays in BF16 (placebo_seed=0). On OLMoE that module is the layer-14 expert-10 down-projection ($1.000\times$ the budget); on Qwen, the layer-19 expert-45 down-projection ($0.978\times$). The control matches parameter count and nothing else. The routers are 16 (24) small modules read by every token in every layer; the placebo is one expert down-projection in one layer, on the compute path for 11.4% (5.0%) of routing tokens. Counting token $\times$ layer events, the routers are exposed about $140\times$ and $477\times$ more. The control rules out “any high-precision parameters help.”

Four-cell attribution of the residual. Even a BF16 router can misroute if the hidden state arriving at it has already drifted through quantized attention and experts. Table 1 crosses gold versus quantized router weights with gold versus quantized incoming activations. The four logit sets are offline matrix multiplications on activations captured during the main runs. Cell C is MIXED: the same attention and expert weights as UNIFORM, with no feedback from perturbed routing. Cell D pairs quantized router weights with those same MIXED activations, so it is **not** UNIFORM. In a real UNIFORM run a rounded router sends a token to different experts, those experts write different values

	gold W	quantized W
gold X	A: reference	B: weight error
quantized X	C: activation drift (=MIXED)	D: both, no feedback ($\neq$ UNIFORM)

Table 1: The four-cell design. All four cells are logits = XW^T on captured activations. Metrics are scored against cell A.

into the residual stream, and every later router therefore sees a different input. Cell D holds its inputs fixed, so it measures the rounded router weights alone and omits that feedback. We score every cell against the rebuilt reference A, and we quote only one share: among tokens flipped by exactly one mechanism, the fraction flipped by the router’s own weights, $B/(B+C)$. The two mechanisms are not additive (Section 6), so we do not partition D.

5 Experiments

Setup. We evaluate OLMoE-1B-7B-0924 (Muenighoff et al., 2024) and Qwen1.5-MoE-A2.7B (Qwen Team, 2024; Bai et al., 2023) on the WikiText-2 test split (Merity et al., 2017): 288,730 / 299,078 tokens, of which 288,448 / 298,785 are scored under a 1024-token window with stride 1024. Routing metrics use a seeded subset of 128 sequences $\times$ 256 tokens (32,768 tokens per layer), with 95% percentile bootstrap intervals from 200 sequence-level resamples (seed=42). We sweep INT8, INT4 and INT3. There is no calibration set and no second corpus.

Router protection helps, on both architectures. Table 2 is the main result. MIXED beats UNIFORM on routing KL and top-1 flip rate at every bit-width on both models. The 95% confidence intervals never overlap (Tables 3 and 4). The effect grows as precision drops: on OLMoE, routing KL falls from 0.0085 to 0.0064 at INT8, 0.0286 to 0.0090 at INT4, and 0.1049 to 0.0385 at INT3. Qwen repeats the pattern. Perplexity moves in the same direction at INT8 and INT4. Qwen INT3 is saturation, perplexities of 9555 and 1236 against a GOLD of 7.97, and we do not treat those ratios as a quality result. Routing and collapse metrics at INT3 stay in range and remain monotone.

The effect is router-specific. PLACEBO reproduces UNIFORM on every reported metric, at all three bit-widths, on both models (Table 2). Protecting a random 0.03% of parameters is not what

Model	Policy	Bits	PPL	Routing KL	Top-1 flip
OLMoE	GOLD	BF16	8.36	0.0000	0.00%
	UNIFORM / PLACEBO	INT8	9.39 / 9.39	0.0085 / 0.0085	18.01% / 18.01%
	MIXED	INT8	9.32	0.0064	13.76%
	UNIFORM / PLACEBO	INT4	10.49 / 10.49	0.0286 / 0.0286	33.01% / 33.01%
	MIXED	INT4	9.69	0.0090	16.86%
	UNIFORM / PLACEBO	INT3	30.23 / 30.23	0.1049 / 0.1049	52.52% / 52.52%
	MIXED	INT3	18.64	0.0385	32.15%
Qwen	GOLD	BF16	7.97	0.0000	0.00%
	UNIFORM / PLACEBO	INT8	9.35 / 9.35	0.0235 / 0.0235	15.65% / 15.65%
	MIXED	INT8	9.28	0.0175	12.75%
	UNIFORM / PLACEBO	INT4	12.01 / 12.01	0.0861 / 0.0861	30.62% / 30.62%
	MIXED	INT4	11.24	0.0355	17.90%
	UNIFORM / PLACEBO	INT3	9555 / 9563	0.4901 / 0.4901	72.48% / 72.48%
	MIXED	INT3	1236	0.3147	55.69%

Table 2: Main results. GOLD is the BF16 upper bound. PLACEBO is printed beside UNIFORM because the two agree to several decimals; that is the control passing. Top- k Jaccard distance tracks the flip rate in every cell (Table 11). Qwen INT3 is saturated; perplexity claims in the text use INT8 and INT4 only.

Model	Bits	UNIFORM	MIXED
OLMoE	INT8	18.01 [17.81,18.18]	13.76 [13.61,13.90]
OLMoE	INT4	33.01 [32.71,33.31]	16.86 [16.71,16.99]
OLMoE	INT3	52.52 [52.21,52.83]	32.15 [31.86,32.39]
Qwen	INT8	15.65 [15.47,15.80]	12.75 [12.58,12.90]
Qwen	INT4	30.62 [30.44,30.82]	17.90 [17.69,18.13]
Qwen	INT3	72.48 [72.11,72.85]	55.69 [55.41,55.99]

Table 3: Top-1 flip rate (%). 95% percentile intervals from 200 sequence-level resamples. Every UNIFORM–MIXED pair is disjoint.

Model	Bits	UNIFORM	MIXED
OLMoE	INT8	8.54 [8.33,8.76]	6.36 [6.16,6.55]
OLMoE	INT4	28.58 [28.10,29.10]	9.01 [8.73,9.30]
OLMoE	INT3	104.9 [102.4,107.5]	38.46 [36.77,40.16]
Qwen	INT8	23.47 [22.86,24.04]	17.48 [16.89,18.04]
Qwen	INT4	86.12 [84.86,87.38]	35.49 [34.59,36.46]
Qwen	INT3	490.1 [484.6,496.5]	314.7 [310.6,319.0]

Table 4: Routing KL ($\times 10^{-3}$), same intervals. Reductions are 25.5%, 68.5%, 63.3% on OLMoE and 25.5%, 58.8%, 35.8% on Qwen.

helps.

Follow-up: attention as well, not instead. Figure 1 includes a later series, ATTENTION, that also leaves every attention projection in BF16 (Kim et al., 2023). Attention is 3.88% of OLMoE and 2.81% of Qwen, about $128\times$ the router budget, so a better perplexity number is the expected tradeoff, not a rival result. The run is an addition to router protection as a design idea, not a replacement for it: the routers stay quantized, and the extra precision is spent upstream. At INT4, that extra budget recovers only 25% / 34% of MIXED’s routing-KL reduction, which is consistent with the routing gain belonging to the routers. On perplexity the larger island helps more, as it should. We expect leaving both the routers *and* attention in BF16 to improve further still, but that is a different tradeoff, and we leave it to a paper whose purpose is that mix.

Expert collapse, resolved per layer. Pooled usage entropy in Table 2’s source files sits at 0.99 everywhere and hides the only real collapse in the sweep. Per layer, Qwen INT3 under UNIFORM starves 222 of 1,440 expert slots (15.4%) and

leaves a fully unused expert in five layers. MIXED starves 55 slots (3.8%) and leaves no unused expert anywhere. The ATTENTION follow-up lands nearby (74 slots, also no unused expert). On OLMoE nothing collapses at any width: 6/1,024 starved slots under UNIFORM INT3 versus 9/1,024 under MIXED, against a GOLD baseline of 17/1,024. That is a boundary condition, not a null we bury. Table 10 and Figure 2 give the per-layer view.

Where the sweep stops being informative. Qwen INT3 is destroyed under every quantized policy (top-1 agreement with GOLD output falls to 6.6% / 12.8%). The ranking among ruined models is not a quality claim. We use INT3 for routing and collapse, which stay finite and ordered, and we make perplexity claims at INT8 and INT4.

6 Attribution Analysis

The rebuild reproduces the runs. Cell A, rebuilt from stored activations and weights, matches the routing decisions the GOLD run recorded at 99.66% mean top-1 agreement on OLMoE (worst layer 99.46%) and 99.37% on Qwen (worst

Model	Bits	B	C	D	$B/(B+C)$
OLMoE	INT8	10.4	13.5	17.5	43.4 _[42.7,44.1]
OLMoE	INT4	27.4	16.6	31.4	62.2 _[61.7,62.8]
OLMoE	INT3	43.8	31.7	50.0	58.0 _[57.6,58.4]
Qwen	INT8	8.8	13.0	15.7	40.3 _[39.8,40.9]
Qwen	INT4	25.0	18.1	30.3	58.0 _[57.5,58.5]
Qwen	INT3*	45.8	55.1	68.6	45.4 _[45.1,45.6]

Table 5: Attribution of top-1 flips (%), 10,000 bootstrap replicates. B, C and D follow Table 1. $B/(B+C)$ is the share of single-mechanism flips due to the router’s own weights. The residual $1 - B/D - C/D$ is -36% to -51% in every row, so the shares of D do not partition the drift. *Saturated.

layer 98.49%). That floor is reconstruction noise: fp16-stored activations, recomputed in CPU fp32, against a BF16 GPU forward pass. Every difference in Table 5 is well above it. Cell C also matches the MIXED flip rates measured in Part 1 (OLMoE INT4: 16.62% rebuilt vs. 16.86% recorded). Cell D falls a little short of UNIFORM, as it must: OLMoE INT3 is 50.01% versus 52.52%; Qwen INT3 is 68.58% versus 72.48%. The gap is the routing-feedback term that D omits.

The dominant mechanism reverses between INT8 and INT4. Of the flips caused by exactly one mechanism, the router’s own weights account for 43.4% / 40.3% at INT8 (OLMoE / Qwen) and 62.2% / 58.0% at INT4 (Table 5). Above half, router protection removes the larger source. That crossover is the same on both architectures, and it is the same on routing KL (Table 12). At INT8 the cheap safeguard is fighting the smaller fire. At INT4 it is fighting the larger one. We do not read Qwen INT3 (45.4%) as activation dominance returning: that cell is the saturated regime of Section 5.

What the share does not say. The interaction residual $1 - B/D - C/D$ is large and negative everywhere (-36% to -51%). The two mechanisms flip overlapping token sets: a token that either source alone would have flipped is counted once in D and twice across B and C. B/D and C/D are therefore not shares of anything that sums to 100%, and we do not write an “ $X\% / (100-X)\%$ ” split.

7 Discussion

Necessary but not sufficient. Leaving the routers in BF16 is a free, specific way to keep routing close to the unquantized model. It is free because the routers are 0.02–0.03% of the weights.

It is specific because a parameter-count-matched placebo does nothing. It is not sufficient for end-task quality, because a large share of the remaining routing error, and most of the perplexity error, comes from the residual stream, not from the router matrix. A later run that also protects attention improves perplexity, as expected from its size; that is extra precision on top of the router question, not an alternative answer to it.

Adjudicating the disagreement. The literature looks inconsistent only if calibration is ignored. EAQuant’s ablation shows that a more precise router helps, on a calibrated W+A baseline, scored on perplexity (Fu et al., 2025). RouteQuant’s headline says freezing the router is not enough, and their own no-alignment split says the opposite in the calibration-free case (Fang and Huang, 2026). We measure that case on routing. Protecting the router is the right default when no data will be spent on it. It does not replace a method that realigns routing with calibration (Park et al., 2026; Hyeon and Do, 2026; Fang and Huang, 2026).

Limitations. The evaluation is WikiText-2, so the reported magnitudes are in-domain. Perplexity is a point estimate. PLACEBO is a single draw, matched on parameter count rather than exposure, and every run uses one seed. We do not report memory or speed: these GPUs unpack the integer weights back to floats before multiplying, so a savings number would not be a real integer-kernel result, and that was not the question we set out to answer. DeepSeek-style routers, whose gate is a bare parameter rather than an `nn.Linear`, perform differently and we did not evaluate their behaviour. (Dai et al., 2024).

8 Conclusion

We asked whether, under calibration-free weight-only quantization, leaving only the MoE routers in BF16 reduces routing drift and perplexity loss. It reduces routing drift, cheaply and specifically, on two architectures and three bit-widths. The residual error switches source between INT8 and INT4, so a full-precision receptionist still misroutes if the chart on her desk is already wrong. Spending more precision on attention as well is a separate, larger tradeoff, in addition, not instead.

AI Disclosure and Reflection

We used Cursor (Grok and related coding agents) to support repository setup, quantization and evaluation code, offline analysis scripts, literature notes, and drafting this report. Every number in the tables was transcribed from the run artifacts and checked against the corresponding JSON. We did not accept generated claims about prior work without reading the papers. The research question, the experimental matrix, the decision to treat the attribution as a mechanism analysis rather than a fallback, and the wording of the controls are ours. The tool sped up implementation and bookkeeping; it did not choose what to measure or what the results mean.

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	OLMoE	Qwen
Layers / experts / k	16 / 64 / 8	24 / 60 / 4
Total params	6.92B	14.32B
Router params	2.10M	2.95M
Router share	0.0303%	0.0206%

Table 6: Architectures, from `inspect_model.py`. Checkpoints: OLMoE-1B-7B-0924 (Muennighoff et al., 2024) and Qwen1.5-MoE-A2.7B (Qwen Team, 2024; Bai et al., 2023).

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, et al. 2020. [Transformers: State-of-the-art natural language processing](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45.

Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. 2023. [SmoothQuant: Accurate and efficient post-training quantization for large language models](#). In *Proceedings of the 40th International Conference on Machine Learning*.

A Experimental setup

Models. Table 6 records the two checkpoints. Both expose the router as `nn.Linear` named `mlp.gate`. We pin transformers to 4.x so experts remain individual linear modules rather than a fused 3D stack, which a walker that only swaps `nn.Linear` would silently skip.

Data and protocol. WikiText-2 raw test, documents joined with a blank line and tokenized as one stream. There is no train/test leakage: scales come from the weights, never from activations, and no hyperparameter is selected on the evaluation tokens. Routing metrics and perplexity share that stream; they are not independent confirmations of the same effect. The routing subset is `default_rng(42).choice` over disjoint 256-token blocks. Router inputs for attribution are stored every eighth token.

Hardware and reproducibility. Runs used NVIDIA GeForce RTX 2080 Ti cards (compute 7.5), three for OLMoE and five for Qwen, with torch 2.9.1, transformers 4.57.6 and torchao 0.17.0 (Or et al., 2025; Wolf et al., 2020). Every results file records the git commit, package versions, GPU type and a SHA-256 fingerprint of the token ids. A candidate whose corpus, seed or layout disagrees with GOLD is rejected rather than scored.

Policy	OLMoE	Qwen
GOLD	0	0
UNIFORM	3152	4536
MIXED	3136	4512
PLACEBO	3151	4535
ATTENTION	3088	4440

Table 7: Modules quantized per policy, identical at every bit-width. MIXED quantizes exactly 16 (resp. 24) fewer modules than UNIFORM, and those modules are the routers. PLACEBO spares one extra module. The ATTENTION follow-up, listed for completeness, spares 64 (resp. 96) attention projections and keeps every router quantized. GOLD self-comparison is exactly zero KL and zero flips on both models.

Model	Run	Lev.	Cap	= GOLD
OLMoE	GOLD / MIXED	1161	,	16/16
OLMoE	unif./plac. INT8	60	256	0/16
OLMoE	INT4 quantized	16	16	0/16
OLMoE	INT3 quantized	8	8	0/16
Qwen	GOLD / MIXED	1107	,	24/24
Qwen	unif./plac. INT8	46	256	0/24
Qwen	INT4 quantized	16	16	0/24
Qwen	INT3 quantized	8	8	0/24

Table 8: Offline bit-identity check on the !!!!!!!!!!!!!!! dequantized router weights each run stored. **Levels** is the largest number of distinct values in any output channel. INT8 falls short of 256 because a per-channel scale of $\max|w|/127$ on roughly Gaussian weights never reaches the extreme codes. All listed runs pass. `scripts/verify_offline.py`.

Deviations from the approved proposal. The proposal named Qwen/Qwen2.5-MoE-1.4B-A14B, which does not exist and is internally contradictory. We substituted OLMoE-1B-7B, one of the three models EAQuant benchmarks (Fu et al., 2025), and Qwen1.5-MoE-A2.7B as a second architecture. We swept INT8/4/3 rather than INT8 alone, because INT8 is nearly lossless and would have left the comparison inside the noise. We evaluated WikiText-2 only, not C4. Part 2 was described as a fallback for a null Part 1; Part 1 came out positive and we ran the attribution anyway, as a mechanism analysis. The proposal’s contingency of scaling routing scores by activation size was not built.

	OLMoE	Qwen
Module	L14 E10 down_proj	L19 E45 down_proj
Params vs. budget	1.000×	0.978×
Tokens on path	11.43%	5.04%
Exposure gap	140×	477×

Table 9: What PLACEBO protected at placebo_seed=0. One expert down-projection per model, re-derived from a meta-device skeleton. The selection rule stops once the accumulated module is within 25% of the router budget; on both checkpoints the first eligible draw already is.

Model	Policy	Starved	Unused	KL
OLMoE	GOLD	17/1024	0	0.131
OLMoE	UNIFORM INT3	6/1024	0	0.163
OLMoE	MIXED INT3	9/1024	0	0.141
OLMoE	ATTENTION INT3	8/1024	0	0.142
Qwen	GOLD	0/1440	0	0.047
Qwen	UNIFORM INT3	222/1440	5	0.572
Qwen	MIXED INT3	55/1440	0	0.269
Qwen	PLACEBO INT3	220/1440	5	0.572
Qwen	ATTENTION INT3	74/1440	0	0.346

Table 10: Expert collapse, resolved per layer. A slot is starved when an expert receives under a tenth of its fair share in its own layer. Starved counts use all 32,768 routing tokens; load KL is rebuilt from the strided capture, with GOLD measured on the same positions. OLMoE does not collapse; Qwen INT3 does, and MIXED mitigates it.

B Correctness gates and offline verification

C Expert collapse: per-layer detail

D Additional policy comparisons

E Attribution: mechanism detail

F Reproduction

The accompanying repository builds the policies as torchao FqnToConfig dictionaries, runs them through a shared evaluation loop, and refuses to score a candidate whose token fingerprint, topology or router-quantization audit fails. Offline analyses (scripts/analyze.py, attribute.py, collapse.py, verify_offline.py) read only the stored artifacts and need no GPU. The main sweep is:

```
python scripts/run.py \
  --config configs/olmoe.yaml \
  --policies gold uniform mixed placebo \
  --bits 8 4 3
python scripts/analyze.py \
```

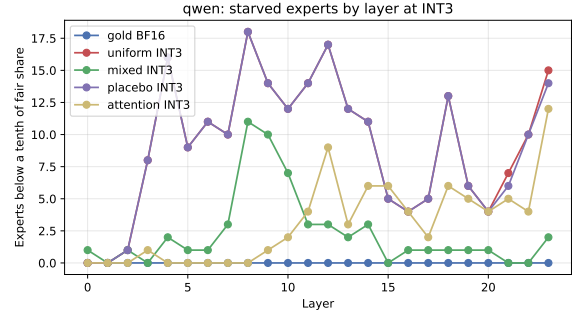


Figure 2: Starved experts by layer on Qwen at INT3, the one cell where collapse occurs. UNIFORM and PLACEBO coincide; MIXED keeps every expert alive, as does the later ATTENTION series. Damage concentrates in the mid-to-late layers, which pooled entropy averaged away. The OLMoE counterpart is flat at every depth and bit-width.

Model	Bits	UNIFORM	MIXED	ATTENTION
OLMoE	INT8	0.145	0.128	0.105
OLMoE	INT4	0.236	0.149	0.206
OLMoE	INT3	0.434	0.277	0.363
Qwen	INT8	0.212	0.185	,
Qwen	INT4	0.356	0.244	0.321
Qwen	INT3	0.744	0.617	0.635

Table 11: Top- k Jaccard distance from the GOLD selected-expert set. GOLD is 0 by construction. The ordering matches the top-1 flip rate. The ATTENTION column is the follow-up series from Figure 1.

```
--results-dir results/olmoe
```

Add attention to -policies for the follow-up series in Figure 1.

Substitute configs/qwen.yaml for the second architecture. Tests (pytest) cover policy construction, metric identities, the four-cell rebuild and the registry collisions that would silently protect the wrong module.

Model	Bits	B	C	D
OLMoE	INT8	0.0018	0.0073	0.0091
OLMoE	INT4	0.0175	0.0099	0.0272
OLMoE	INT3	0.0586	0.0393	0.0929
Qwen	INT8	0.0053	0.0207	0.0260
Qwen	INT4	0.0444	0.0387	0.0836
Qwen	INT3	0.1666	0.3145	0.4540

Table 12: Routing KL by mechanism. The INT8-to-INT4 crossover is the same as in Table 5: $B < C$ at INT8 and $B > C$ at INT4 on both models.